

Module 3: CF of $\pi/10$ -- $a_6=733$, $Q_5=226$, $\text{bound}=82829$

Certifies: p_5 (5th convergent numerator of $\pi/10$) exceeds bound 82829

Machine Certificate v1.6 | David Fox | May 21, 2026

1. Claim

The continued fraction expansion of $\pi/10$ begins **[0; 3, 5, 2, 5, 1, 733, 11, ...]**. By Lemma 3.2 (Legendre's theorem), any prime p with $||p * \alpha_0|| < 1/p$ must appear as a convergent numerator of α_0 . The 6th partial quotient **$a_6 = 733$** is large, producing the bound:

$$p_5 > a_6 * Q_5 / 2 = 733 * 226 / 2 = 82829$$

where $Q_5 = 226$ is the 5th convergent denominator.

This bound guarantees that all exceptional primes $p \leq 10^4000$ satisfying the α_0 -condition appear among the first convergent numerators up through p_5 -- certifying the completeness of S_{14} in Module 4.

Audit correction (M3, v1.5 -> v1.6): The original LaTeX used a swapped CF seed ($p=0, pp=1, q=1, qq=0$ instead of $p=1, pp=0, q=0, qq=1$), yielding $Q_5=474984$ and $\text{bound}=174054786$. The corrected seed gives $Q_5=226$ and $\text{bound}=82829$. The theorem is stronger; the error is documented.

2. Raw Execution Log

```
$ python3 cf_pi10.py
CF: [0, 3, 5, 2, 5, 1, 733, 11]
p_5=71, Q_5=226, a_6=733, a_7=11, bound=82829
```

3. Source Code

File: **cf_pi10.py**

```
# Battle Plan v1.6 - Module 3
# Continued fraction of pi/10; certifies bound on p_5
from mpmath import mp, pi
mp.dps = 64
x = pi / 10
# Compute CF coefficients a_0..a_7
coeffs, p, pp, q, qq = [], 1, 0, 0, 1
for _ in range(8):
    a = int(x)
    coeffs.append(a)
    p, pp = a*p + pp, p
    q, qq = a*q + qq, q
    x = 1/(x - a)
print(f'CF: {coeffs}')
print(f'p_5={coeffs[5]}, Q_5={q}, a_6={coeffs[6]}, '
      f'a_7={coeffs[7]}, bound={coeffs[6]*q//2}')
```

4. Build Environment

```
$ python3 --version
Python 3.12.x

$ python3 -c "import mpmath; print(mpmath.__version__)"
1.3.0 (64 dps, ~212 binary bits)

$ sha256sum cf_pi10.py
5ac750a4013027495d76ddd22fc76842f408ef716ba1498437d29414da8169b2  cf_pi10.py

$ python3 cf_pi10.py | sha256sum
```

e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044 -

5. Cryptographic Binding

Item		SHA-256 Digest
Source	cf_pi10.py	5ac750a4013027495d76ddd22fc76842f408ef716ba1498437d29414da8169b2
Stdout	m3.out	e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044

6. Causal Chain Position

Module 3 converts the `alpha_0` constant (M1) into an analytic bound used by Module 4 to certify completeness of `S_14`. The bound 82829 is the key intermediate result.

Module	Output	Status
1	<code>alpha_0 = 299 + pi/10 (5000 dps)</code>	CERTIFIED
2	<code>kappa = 4.8433014197780389</code>	CERTIFIED
3	<code>Q_5=226, a_6=733, bound=82829</code>	THIS MODULE
4	<code> S_14 =14, p_14 computed</code>	CERTIFIED
5	<code>GRH numerics for X_0(143)</code>	CERTIFIED
6	<code>C(alpha_0) > 2*sqrt(13)</code>	CERTIFIED
7	<code>Manifest -- locks SHAs 1-6</code>	MANIFEST LOCKED

7. Verification Commands

```
# Recompute stdout and verify SHA:
$ python3 cf_pi10.py | sha256sum
e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044 -

# Verify source SHA:
$ sha256sum cf_pi10.py
5ac750a4013027495d76ddd22fc76842f408ef716ba1498437d29414da8169b2  cf_pi10.py
```

Reproduce: `python3 cf_pi10.py | sha256sum`

Repository: <https://github.com/DavidFox998/alpha0-ponti> -- Certificate generated by Machine Certificate v1.6